

CS218 Final Project Individual Report

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My main responsibilities for this project were designing and implementing the UI, and connecting that UI to our websocket and REST APIs. The UI was intended to be fairly simple, allowing new users to quickly get up and running without a significant learning curve. I also designed the websocket system which allowed users to see each other's actions in real time. The application was initially designed and tested offline prior to deployment on S3, and websockets communicated directly with each other, allowing a test user to open the application in multiple tabs, and each their own duplicate cursor perform the actions they made in the second tab. When we moved to an online hosted solution (Sathaya designed the systems for bringing the application up and down on S3 and Cloudfront) we moved to a Lambda based system for tracking websockets, as well as the REST API used for tracking actions taken in the program. For example, things such as creating a new whiteboard, or drawing within one. Sathaya designed the Lambda functions, which connected to my front end and allowed communication from our program to DynamoDB. James was our head of security, and designed our login system via AWS Cognito.

My first and most poignant takeaway from this project was that the approach we used for designing a cloud system was somewhat backwards. I can't say if I'm speaking for only myself, or my group members as well, but the issue I found myself having when working on this project was one of trying to find something interesting to work on as a project first, *then* trying to find ways to incorporate cloud technology into it. This led to a rather confusing conundrum, where we were working to try to find issues with which to solve with cloud computing, as opposed to being presented with issues which required cloud computing in order to solve. While I do think the solutions we ended up coming up with were interesting, and demonstrate the ability to use cloud computing resources effectively, the process by which we came to those solutions felt somewhat backwards. This is, in part, why our project isn't a novel concept. The priority was to design something we actually found interesting and fun to work on, then finding ways to incorporate cloud computing into the project, over finding a specific problem which was tailored to being solved via cloud computing. If we were to attempt this assignment again, I think I would vote for spending more time on trying to find a specific, cloud computing relevant problem which we could then design a solution around.